Explainable AI for Post-COVID Psychological Profile Prediction

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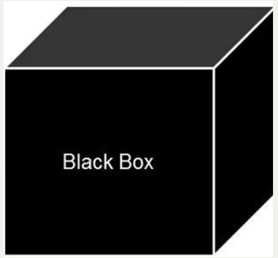
01

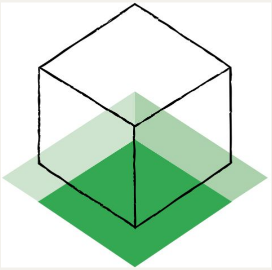
RECAP

Background, significance, and methodology

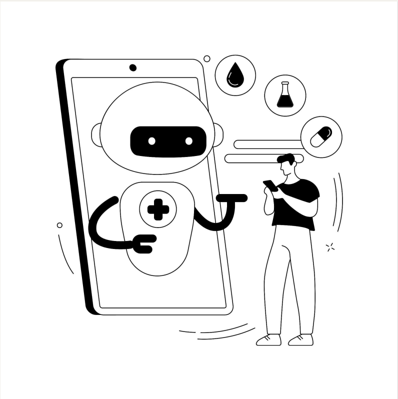
Background & Significance



Lack of transparency in mental health apps



Develop XAI using advanced methods



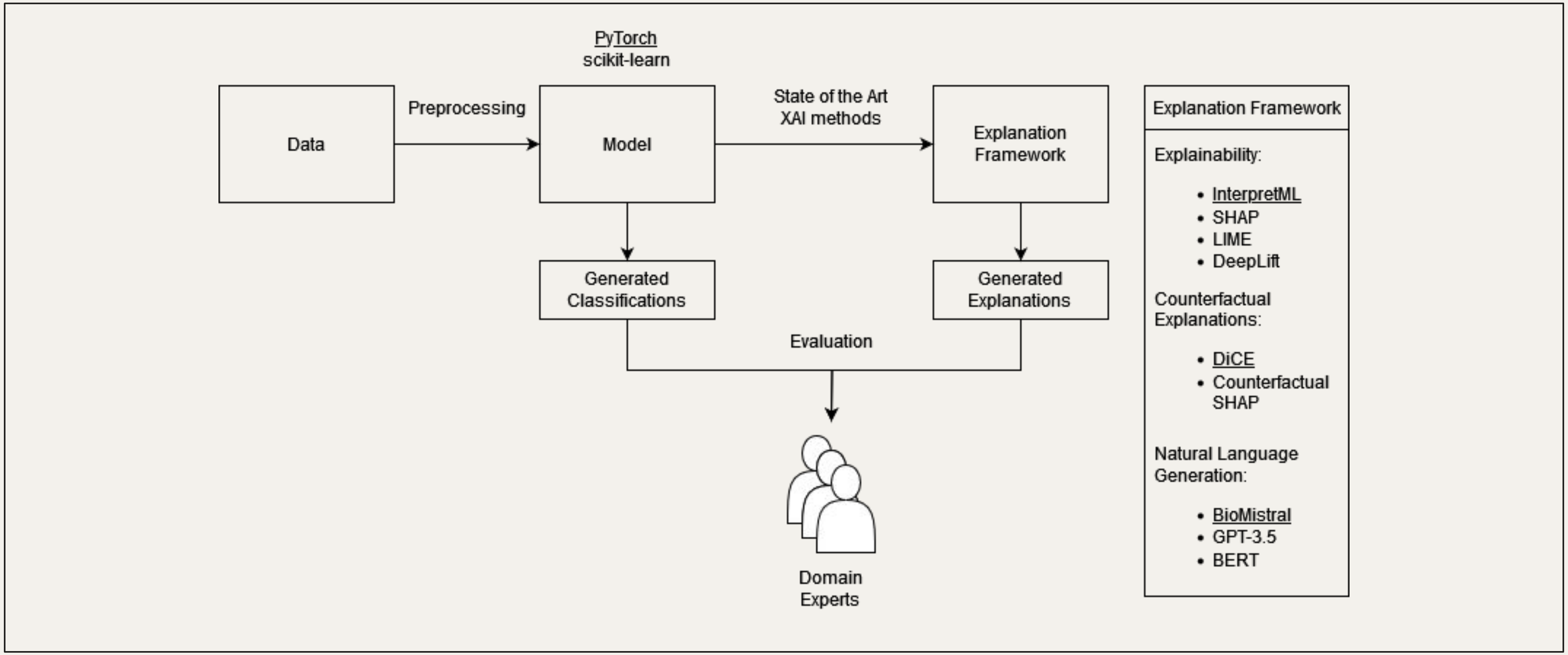
Create clear, contextual explanations for predictions

“

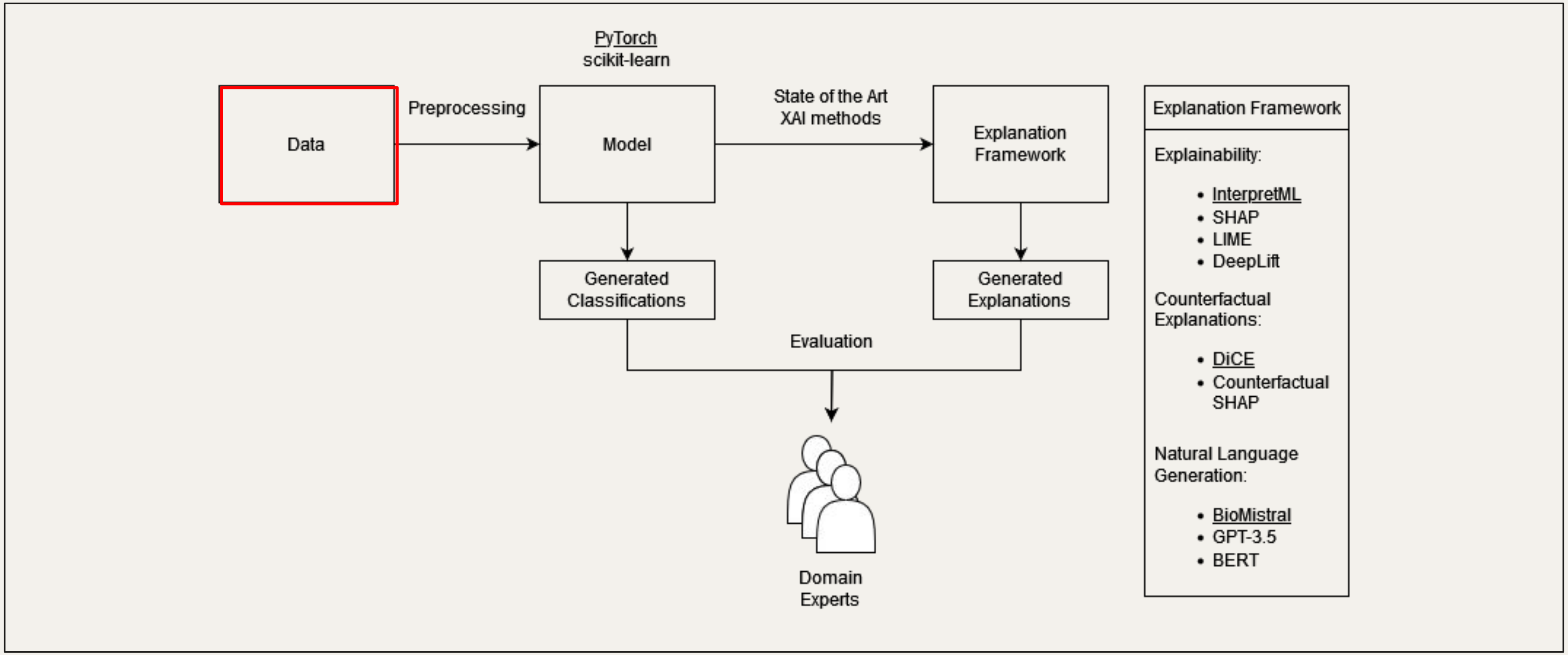
… shedding light on the dynamic nature of mental health outcomes over time and underscoring the importance of continued support and interventions tailored to the changing mental health requirements of this affected population.

— Badinlou et al., 2024

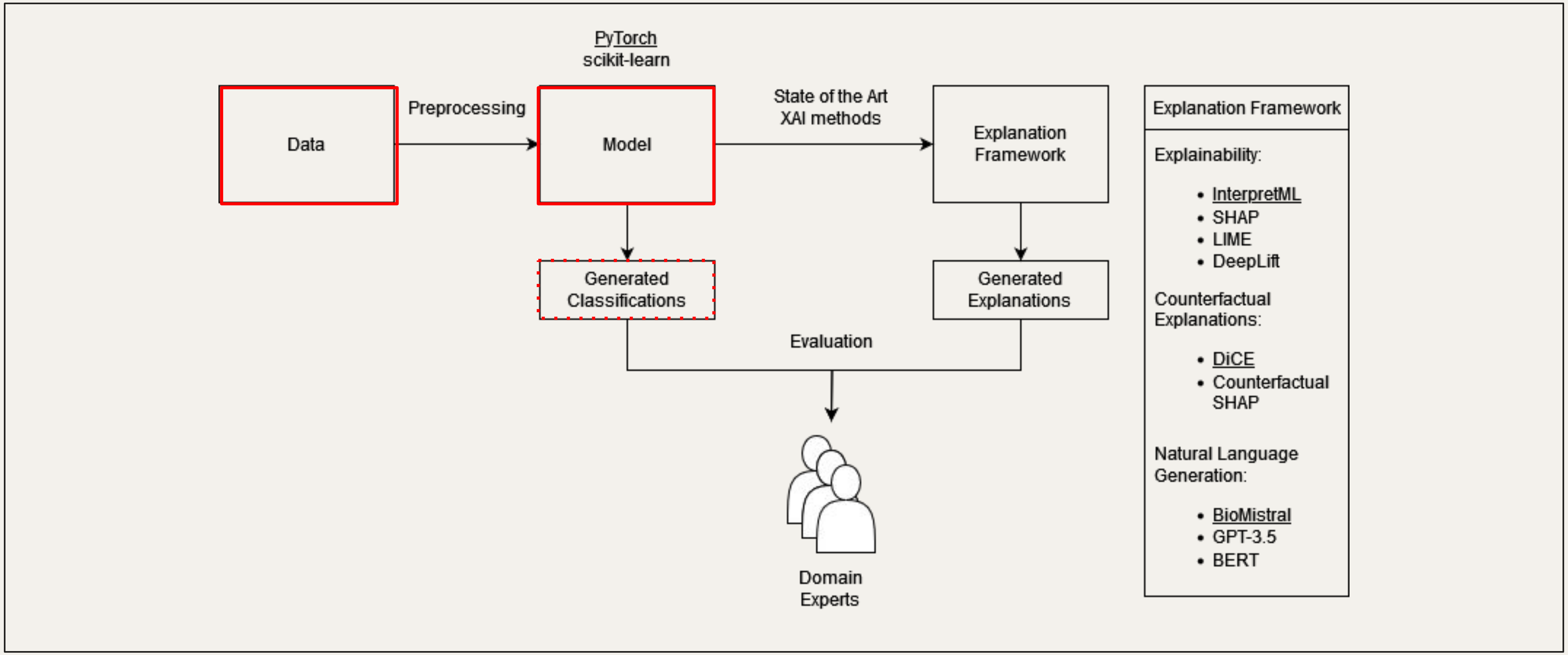
Methodology



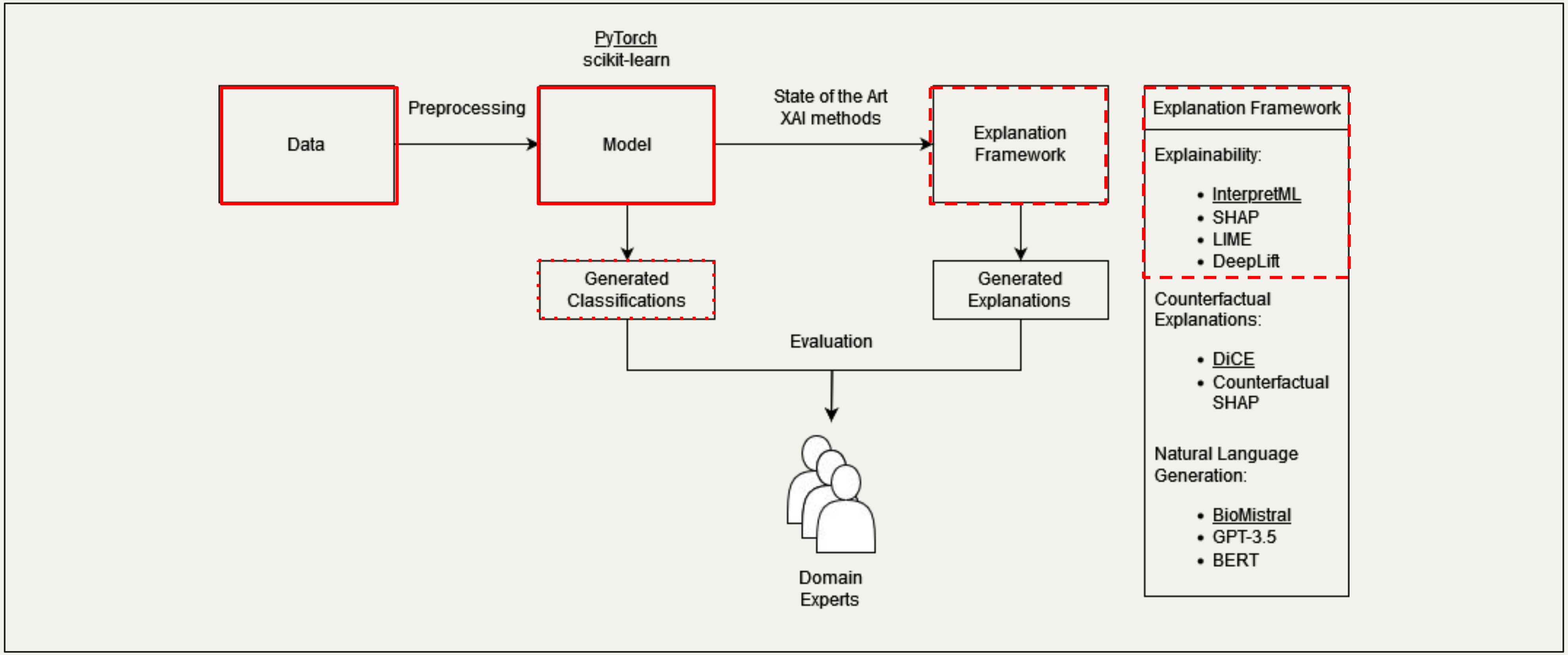
Methodology



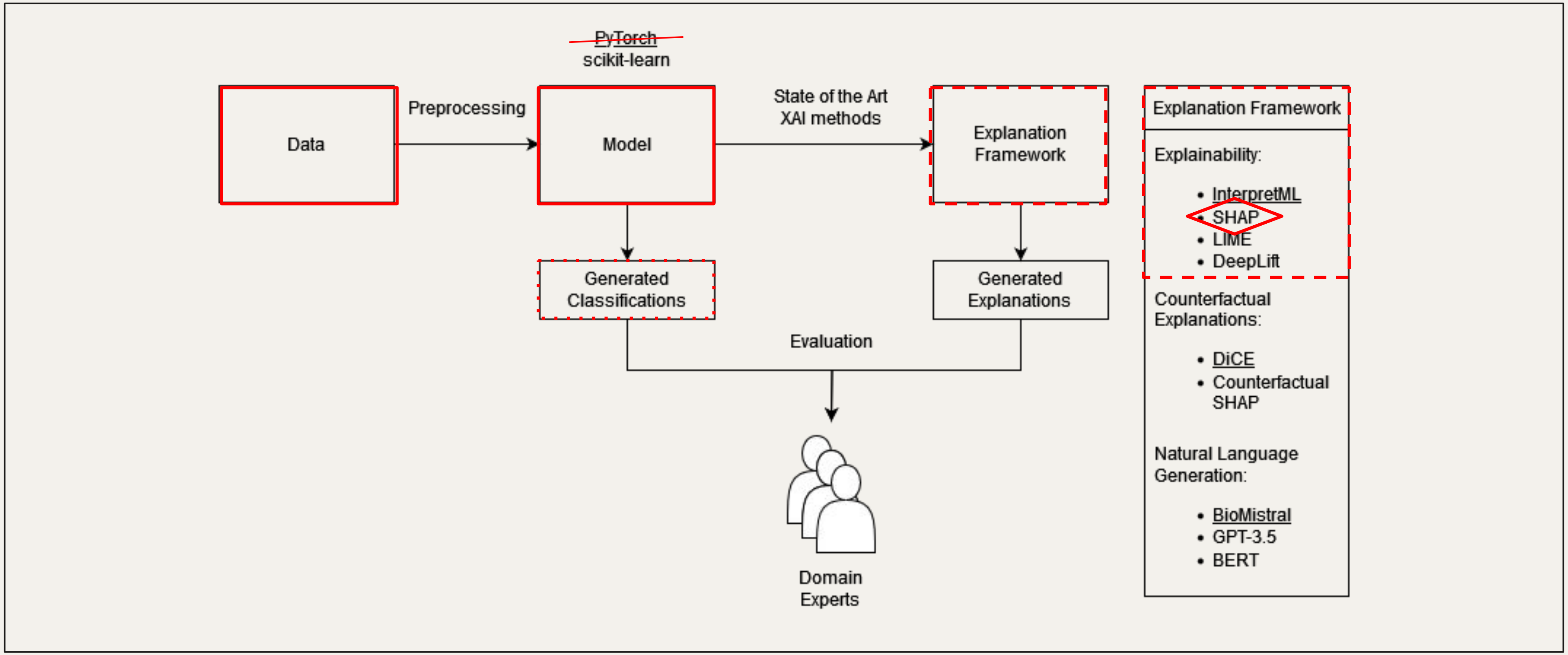
Methodology



Methodology



Methodology



02

DATA

Preprocessing decisions

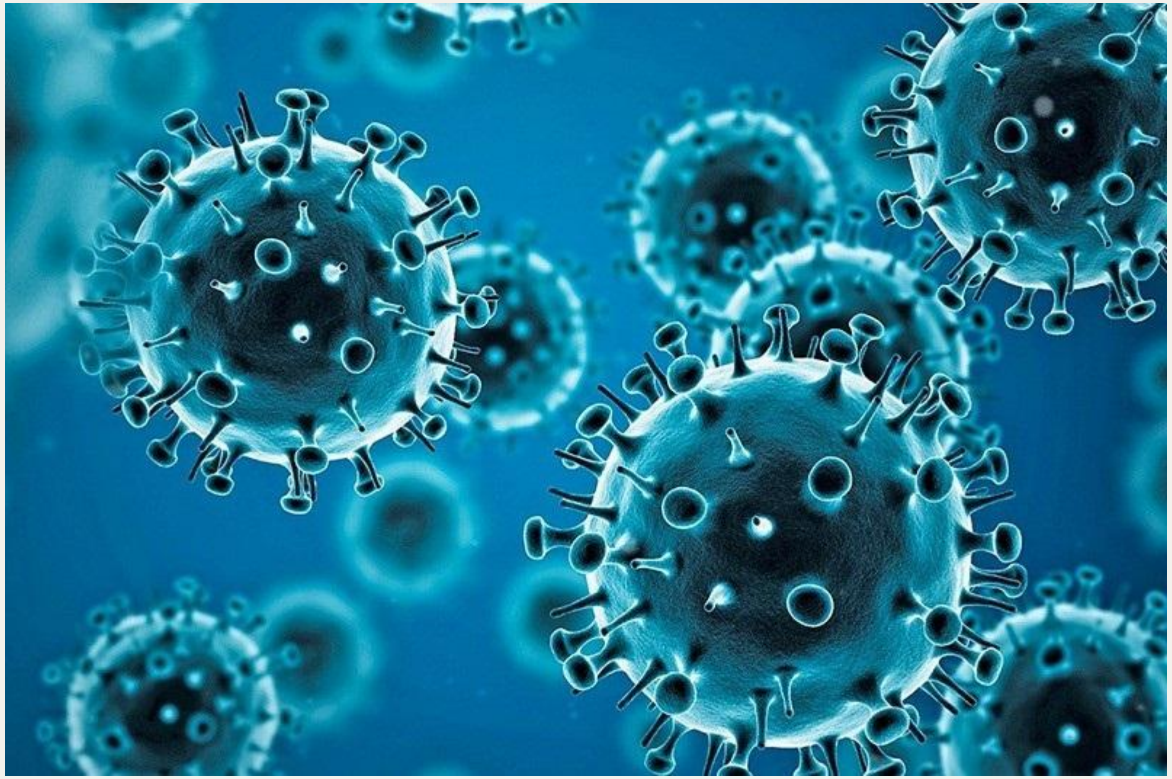
Socio-Demographic Information

5_SEX
5_AGE
5_MARRIED
5_PINCOME
5_JOB
5_STUDENT

COVID-19-Related Information

5_VacNum
5_Med_fam
5_Medfam_covid
5_Current_physical
5_Current_covid19
5_Trust_gov
5_Trust_SM

5_Vaccination_will
5_DeteriorationEconomy
5_DeteriorationInteract
5_CovidAnxiety
5_CovidSleepless



Mental Health Information

5_Current_mental
5_K6_total
5_PHQ9_total
5_GAD7_total
5_SSS8_total
5_PTGI-X(Q7_39_44)

5_SHS_total
5_UCLA_total
5_LSNS6_total
5_Frustration
5_DifficultyLiving
5_DifficultyWork

Coping Strategies Information

5_AUDIT
5_Exercise
5_HealthyDiet
5_FavoriteActivity
5_Interaction_offline
5_Interaction_online

5_PB_Continuous
5_PB_Altruistic
5_PB_Avoidant
5_PB_understanding
5_Optimism
5_HealthySleep

Targets

K6

0-12: Unlikely distress
13-24: Likely distress

PHQ-9

0-4: Minimal depression
5-9: Mild
10-14: Moderate
15-19: Moderately severe
20-27: Severe

GAD-7

0-4: Minimal anxiety
5-9: Mild
10-14: Moderate
15-21: Severe

SSS-8

0-3: Minimal somatic symptom burden
4-7: Low
8-11: Medium
12-15: High
16-32: Very high

UCLA Loneliness Scale

0-19: Minimal loneliness
20-34: Low
35-49: Moderate
50-64: Moderately high
65-80: High

LSNS-6

0: Minimal social engagement
1-59: Intermediate social engagement
60: High social engagement

SHS

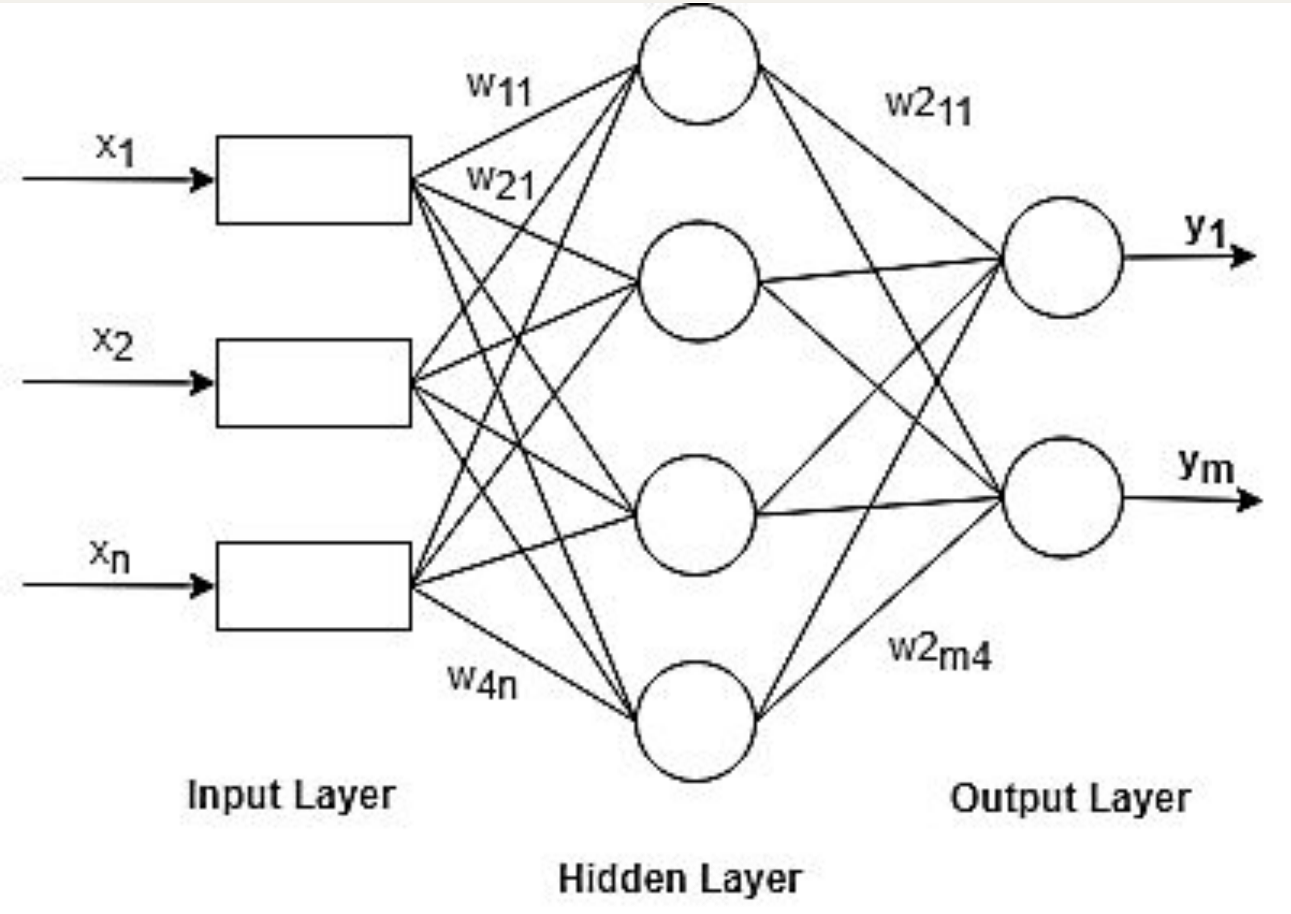
1: Low happiness
2-6: Intermediate happiness
7: High happiness

03

MODELS

Models used to build the classifiers

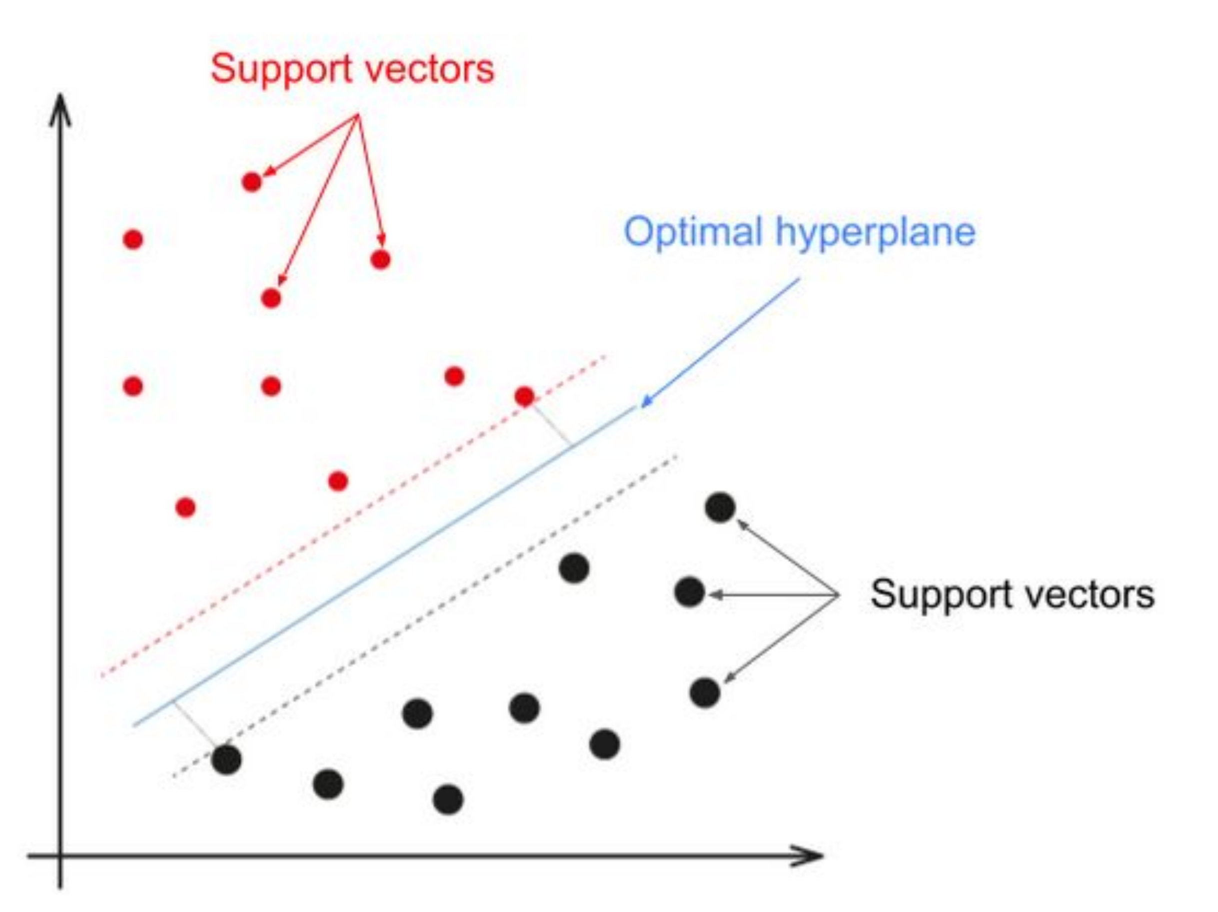
MLPClassifier



Hyperparamters

```
hidden_layer_sizes=(100,),
activation='relu',
solver='adam',
alpha=0.0001,
batch_size='auto',
learning_rate='constant',
learning_rate_init=0.001,
power_t=0.5,
max_iter=200,
shuffle=True,
random_state=None,
tol=0.0001,
verbose=False,
warm_start=False,
momentum=0.9,
nesterovs_momentum=True,
early_stopping=False,
validation_fraction=0.1,
beta_1=0.9, beta_2=0.999, epsilon=1e-08,
n_iter_no_change=10,
max_fun=15000
```

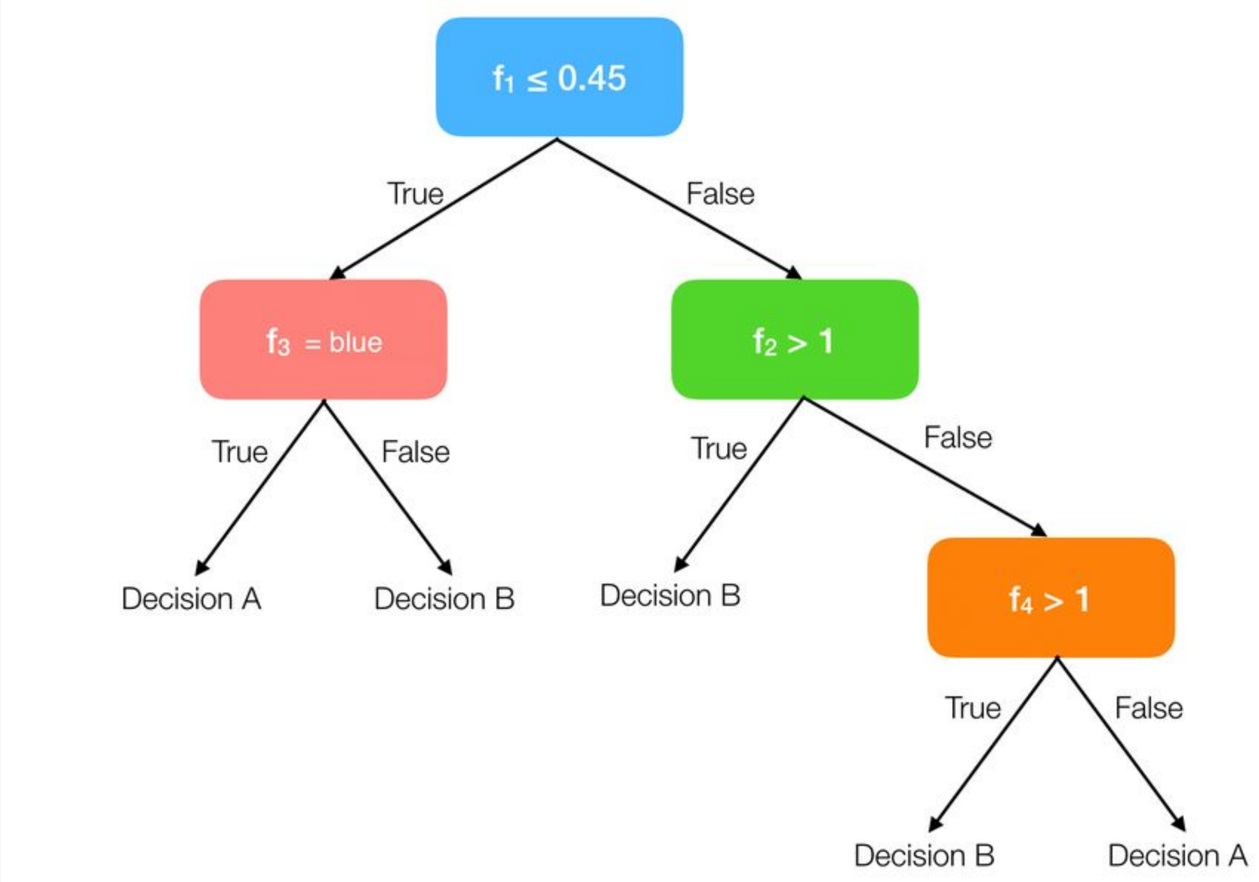
SVM



Hyperparamters

```
C=1.0,
kernel='rbf',
degree=3,
gamma='scale',
coef0=0.0,
shrinking=True, probability=False,
tol=0.001,
cache_size=200, class_weight=None,
verbose=False,
max_iter=-1,
decision_function_shape='ovr',
break_ties=False, random_state=None
```

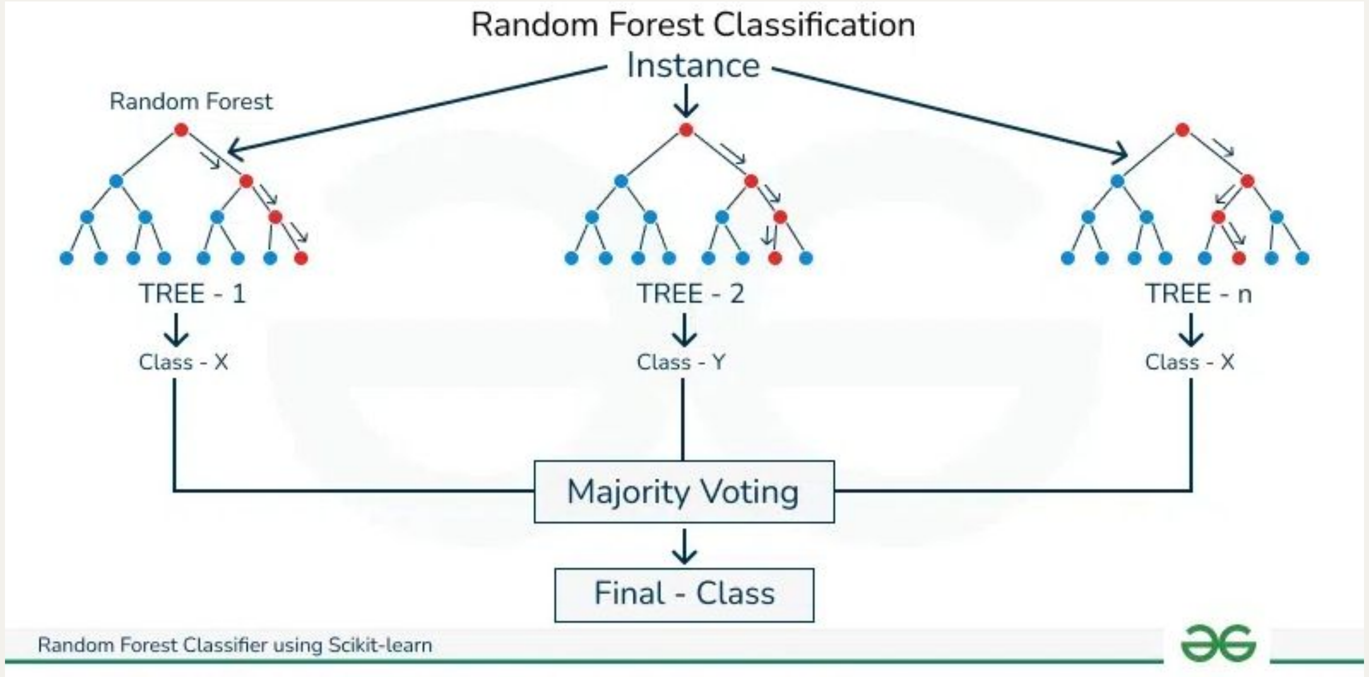
Decision Tree



Hyperparamters

```
criterion='gini', splitter='best',
max_depth=None, min_samples_split=2,
min_samples_leaf=1,
min_weight_fraction_leaf=0.0,
max_features=None, random_state=None,
max_leaf_nodes=None,
min_impurity_decrease=0.0,
class_weight=None,
ccp_alpha=0.0,
monotonic_cst=None
```

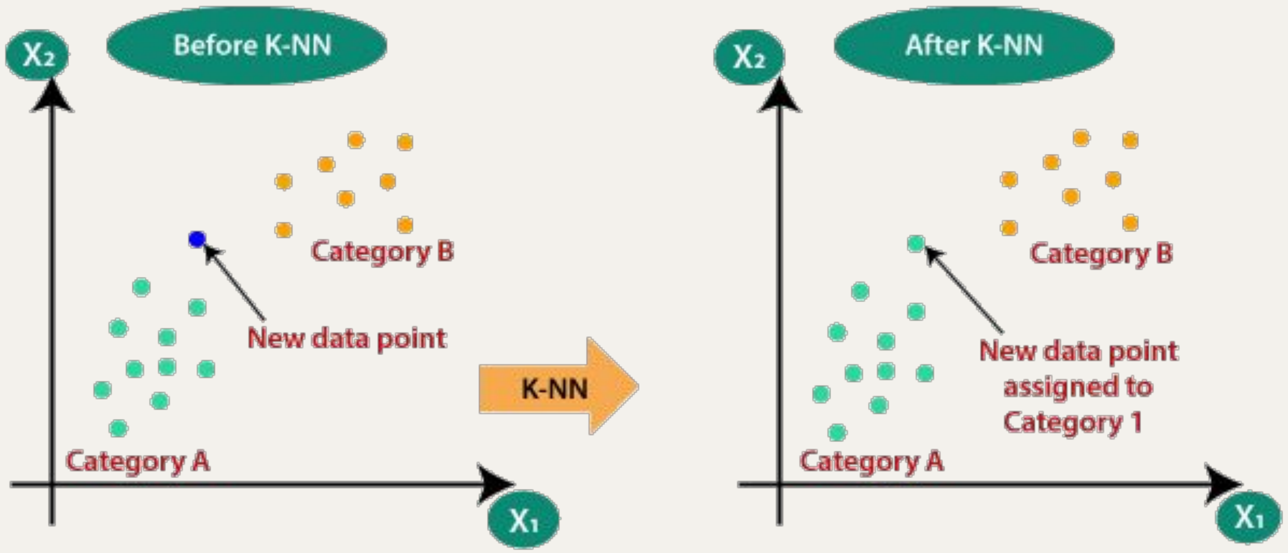
Random Forest Classification



Hyperparamters

```
n_estimators=100, criterion='gini',
max_depth=None, min_samples_split=2,
min_samples_leaf=1,
min_weight_fraction_leaf=0.0,
max_features='sqrt',
max_leaf_nodes=None,
min_impurity_decrease=0.0,
bootstrap=True,
oob_score=False,
n_jobs=None,
random_state=None,
verbose=0,
warm_start=False, class_weight=None,
ccp_alpha=0.0,
max_samples=None, monotonic_cst=None
```

KNNeighbors

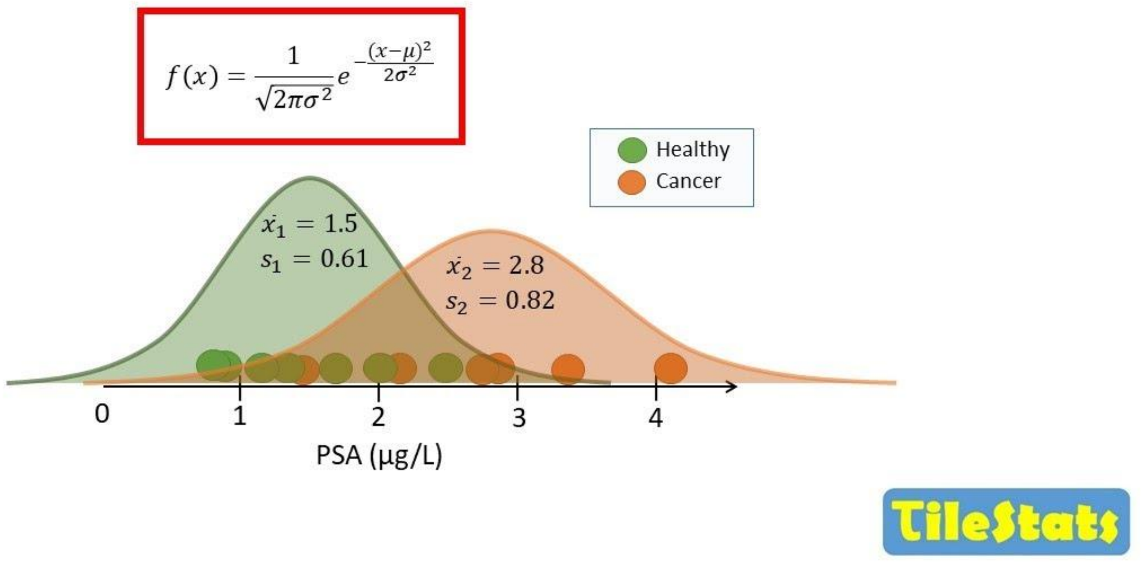


Hyperparamters

```
n_neighbors=5,
weights='uniform', algorithm='auto',
leaf_size=30,
p=2,
metric='minkowski', metric_params=None,
n_jobs=None
```

Gaussian Naive Bayes

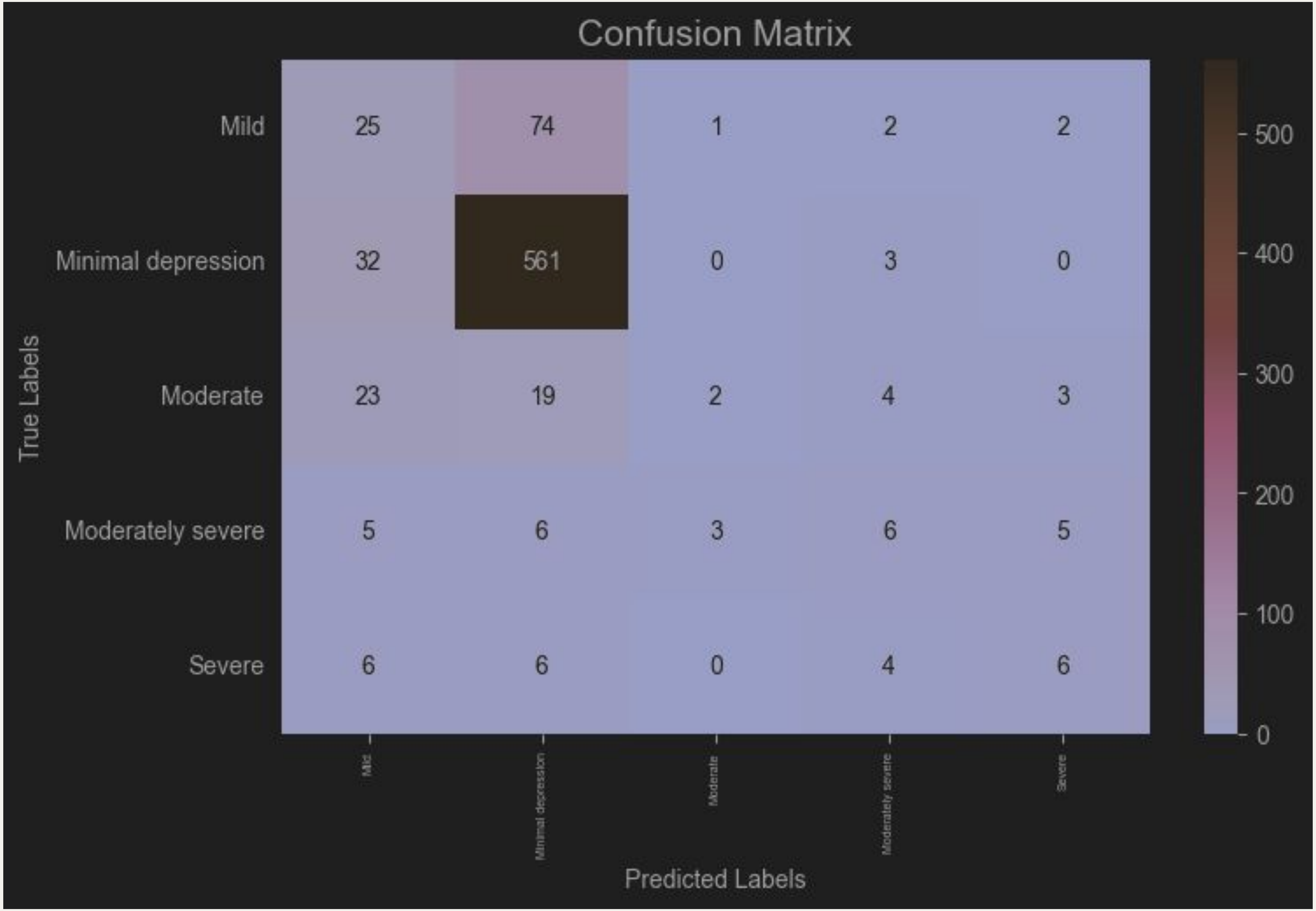
Status	PSA
Cancer	4.1
Cancer	3.4
Cancer	2.9
Cancer	2.8
Cancer	2.7
Cancer	2.1
Cancer	1.6
Healthy	2.5
Healthy	2.0
Healthy	1.7
Healthy	1.4
Healthy	1.2
Healthy	0.9
Healthy	0.8



Hyperparamters

priors=None,
var_smoothing=1e-09

Model Performances Summary



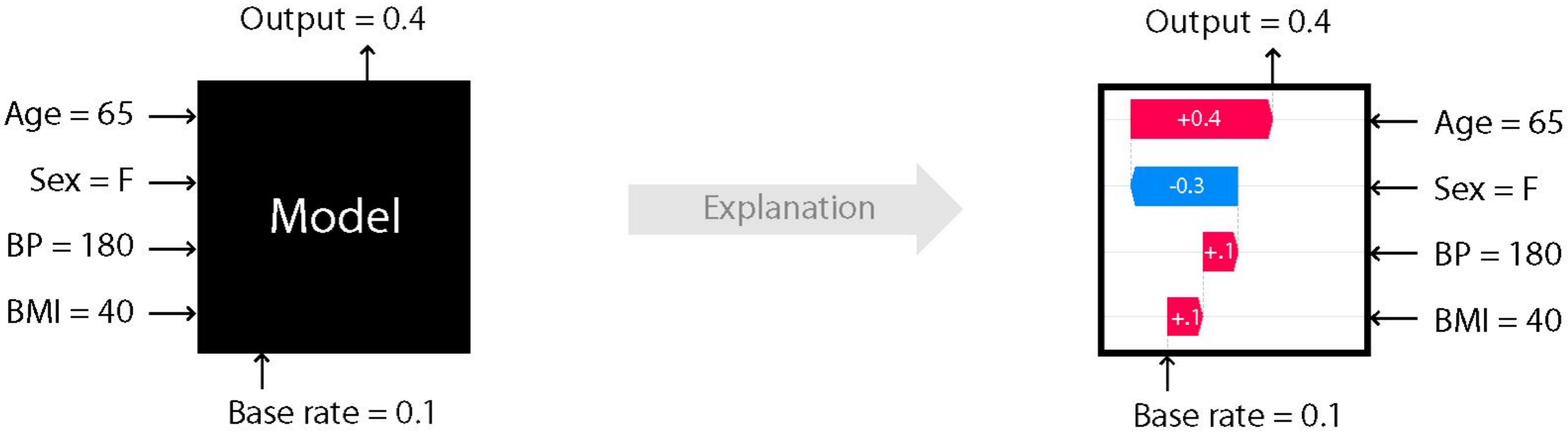
Confusion Matrix
True Labels
Mild
Minimal depression
Moderate
Moderately severe
Severe
25
74
1
2
2
32
561
0
3
0
23
19
2
4
3
5
6
3
6
5
6
6
0
4
6
Mild
Minimal depression
Moderate
Moderately severe
Severe
Predicted Labels
500
400
300
200
100
0

04

EXPLAINABILITY

SHAP & Demo!

SHAP


Output = 0.4
Age = 65
Sex = F
BP = 180
BMI = 40
Model
Base rate = 0.1
Explanation
Output = 0.4
+0.4
-0.3
+.1
+.1
Age = 65
Sex = F
BP = 180
BMI = 40
Base rate = 0.1

05

UPDATED TIMELINE

The plan moving forward

Thank you!

Q&A

Sources

Presentation Template: SlidesMania

Sample Images: Unsplash

Fonts used in this presentation: Poppins Light and **Poppins Bold.**